

Falsification sequence: DVG → DPH → FLH → SDPH

Stage	Hypothesis	Test	Result	Interpretation
DVG	Static directional value-vector geometry (8 cosine/norm features) predicts the downstream winner.	Pooled + within-stratum AUC/correlation on Pythia (n=1,962) and GPT-2 (n=22,477, ~10x the power).	Best CV AUC 0.550 (Pythia) vs. 0.522 for Gamma alone; 4/8 features near-zero despite added power on GPT-2; 0/8 reach the within-stratum effect-size threshold.	Failed as a useful predictor; the larger sample strengthens the negative result rather than rescuing it.
DPH	The downstream winner emerges only through propagation across depth.	Cluster-aware causal trace of 250 stratified cases (100 Pythia, 150 GPT-2); cell-level bootstrap, B=10,000.	Pooled flip rate 32.4% [26.1,39.2]; architecture-difference CI crosses zero (Pythia 38.0% [26.4,52.2] vs. GPT-2 28.7% [21.6,36.0]).	Reversal is substantial but heterogeneous; no universal depth rule; architectures not reliably distinguishable.
FLH	Logit/KL-space (functional) representation resolves the residual-space ambiguity.	Same 250 cases reprojected via the logit lens; paired cluster bootstrap against the DPH result.	$\Delta_{flip} = -0.156$ [-0.231,-0.082]; $\Delta_{unresolved} = +0.392$ [0.294,0.485].	Fewer flips but substantially more unresolved cases; functional space does NOT cleanly resolve the ambiguity.
SDPH	The same downstream block responds differently to an identical perturbation depending on the policy-conditioned base state, explaining the reversals.	n=24 causal swap-intervention audit, finite-difference scale sweep; validation error 0.0.	State-dependence is real and near-linear, but flip/stable median-S ratio = 0.286 (opposite the predicted direction); architecture-inconsistent.	State-dependence CONFIRMED to exist, but FALSIFIED as the explanation for flip status.